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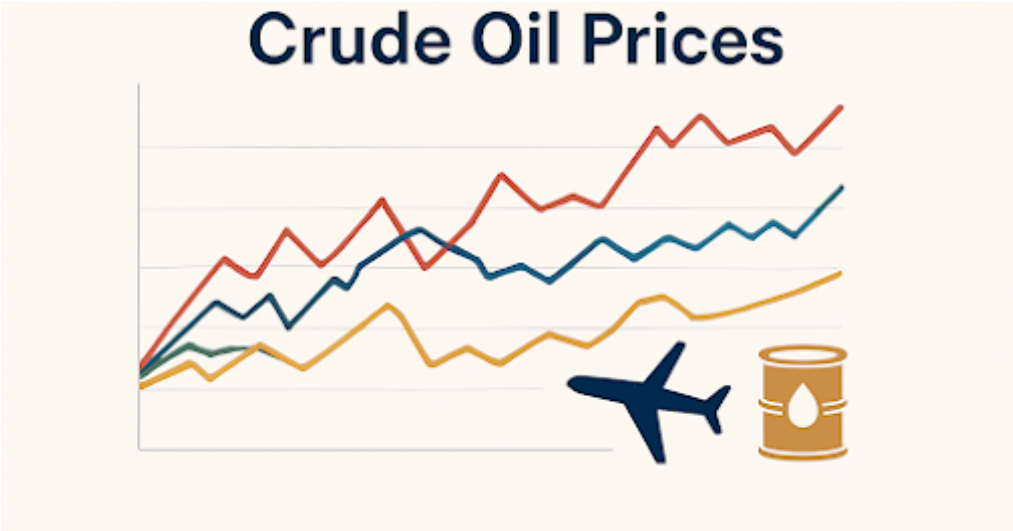


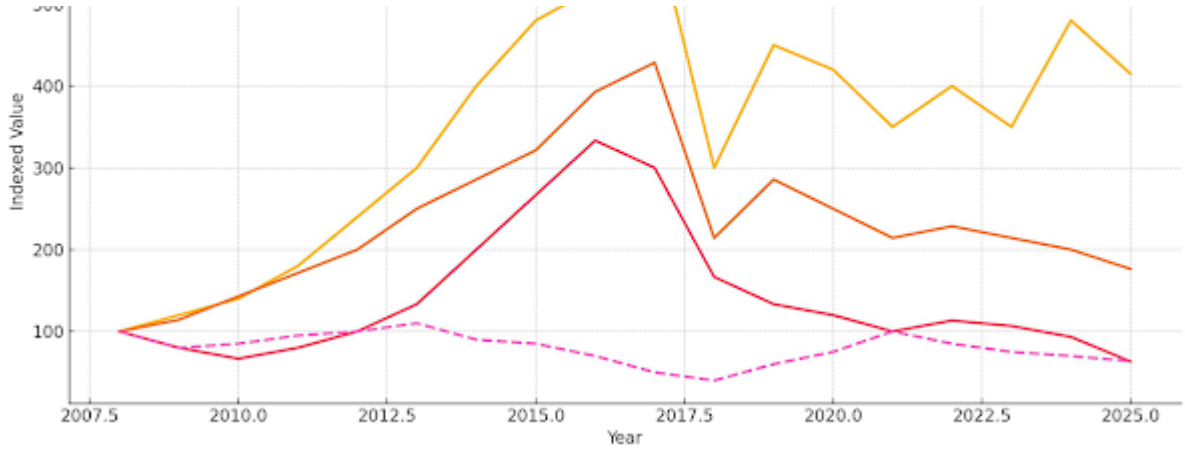
The Correlation Between U.S. Airline Stocks and Crude Oil Prices (2008–2025): A Long-Term Analysis

How Oil Prices Impact Airline Performance

The airline industry is known for its sensitivity to fluctuations in fuel prices. With jet fuel expenses comprising a significant portion of total operating costs—often as high as 30%—any movement in crude oil prices can directly affect an airline's bottom line. Over the past two decades, the volatility in oil markets and the evolving airline business models have added layers of complexity to this relationship.

This article provides a detailed, long-term correlation analysis between three of the largest U.S. airline stocks—**Delta Air Lines (DAL)**, **Southwest Airlines (LUV)**, and **American Airlines (AAL)**—and **WTI crude oil (USOIL)** prices, using weekly data from 2008 to 2025. The goal is to uncover how these airline stocks have historically reacted to oil price fluctuations and what this could mean for future trends.

Darrisman's



Indexing the Data: Establishing a Common Starting Point

To compare trends over time, we indexed all data from a common baseline: the year 2008.

- **Delta Air Lines (DAL)**: Starting price approximately $10
- **Southwest Airlines (LUV)**: Starting price approximately $14
- **American Airlines (AAL)**: Starting price approximately $15
- **WTI Crude Oil (USOIL)**: Starting price approximately $100

This approach allows us to focus on **relative performance**, making it easier to track percentage gains or losses across assets with different absolute price levels.

Delta Air Lines (DAL) and Crude Oil: Moderate Inverse Correlation

Trend Overview

From 2008 to early 2020, Delta Air Lines experienced substantial growth, rising from ~$10 to over $60 before the COVID-19 crash in early 2020. Oil prices, on the other hand, showed a high in 2008, dropped during the financial crisis, then oscillated between $40 and $110 until the pandemic triggered a dramatic collapse.

Post-2020 Rebound and Energy Volatility

2022 before retracing below $70.

Correlation Insight

There is a **moderate inverse correlation** between DAL and USOIL—especially visible during rapid oil rallies, where Delta's margins are squeezed, causing stock pullbacks. However, the correlation is not always perfect, as broader economic conditions and air travel demand also play large roles.

Southwest Airlines (LUV) and Crude Oil: Unique Behavior Amid Volatility

Trend Overview

LUV climbed steadily from ~$14 in 2008 to nearly $60 by 2017–2018. It then experienced prolonged sideways movement followed by sharp declines post-2019. The COVID-19 crisis saw the stock fall below $25 before a brief rebound. As of 2025, the stock sits around $25.

Fuel Hedging Strategy

Southwest is known for its **fuel hedging strategy**, which partially shields it from short-term oil price spikes. This practice helps decouple the LUV stock price from crude oil movements compared to competitors.

Correlation Insight

LUV has demonstrated a **weaker correlation** with crude oil prices due to its strategic hedging. Even during oil price surges, LUV was relatively less impacted—though long-term trends (like post-2022 declines) suggest macroeconomic forces eventually outweigh hedging benefits.

American Airlines (AAL) and Crude Oil: High Sensitivity, Lower Resilience

Trend Overview

to regain momentum, hovering below $20 in recent years.

High Operating Leverage

AAL operates with **higher debt levels and thinner margins**, which magnify its exposure to oil price fluctuations. Spikes in energy prices tend to disproportionately impact AAL's performance, as the airline has fewer buffers to absorb cost shocks.

Correlation Insight

AAL shows the **strongest inverse correlation** with crude oil among the three airlines. During oil rallies, especially from 2021 to 2022, AAL significantly underperformed. This sensitivity makes it a riskier but potentially high-reward investment if oil prices fall.

Crude Oil (USOIL) Behavior: A Volatile Anchor

Trend Overview

From 2008's highs of ~$140, oil fell below $40 during the financial crisis, then oscillated around $80–100 until 2014. A major collapse followed, with oil hitting below $30 in 2016. The 2020 COVID-19 crisis led to a temporary crash below $0 (futures), before prices skyrocketed above $120 in 2022 due to post-pandemic demand and geopolitical tensions.

Post-2022 Trends

After peaking, oil prices gradually declined, stabilizing around $60–70 by 2025. This range appears to be a new normal for the energy market amid increasing global production and softening demand.

Airline	Starting Price (2008)	Correlation to Oil	Key Observations
Delta (DAL)	~$10	Moderate Inverse	Post-2020 recovery; reacts to oil rallies
Southwest (LUV)	~$14	Weak Inverse	Fuel hedging buffers oil impact
American (AAL)	~$15	Strong Inverse	High oil sensitivity; volatile performance

Investor Takeaways

For investors in airline stocks, monitoring WTI crude oil is essential. Spikes in oil prices often precede pullbacks in airline stocks, especially those with high operating leverage.

2. Not All Airlines React Equally

While the industry moves as a group, differences in financial structure and fuel strategy matter. LUV's hedging strategy offers some cushion, while AAL's high debt and limited flexibility amplify risk.

3. Strategic Portfolio Allocation

A diversified airline investment strategy could involve a mix of Delta (as a balanced pick), Southwest (for defensive exposure), and selective entry into AAL (for high-risk, high-reward plays during low oil cycles).

Conclusion : What This Means for Investors

This long-term analysis from 2008 to 2025 reveals that airline stock performance is indeed influenced by crude oil prices—but the degree of correlation varies significantly across carriers. Delta, Southwest, and American Airlines each exhibit different sensitivities based on their fuel management, balance sheet structure, and broader strategic decisions.

For long-term investors, understanding these nuances is critical. By combining technical analysis with macroeconomic awareness, one can make more informed decisions about investing in the airline sector amid volatile oil markets.

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